The Promise, Roadblocks, and Broader Significance of W3C's Attribution Standard

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Despite major progress in privacy technologies and platform controls, the dominant model for data-driven functionality still feels stuck. Modern applications increasingly rely on aggregation – population-level statistics, analytics, and machine-learning model training – rather than raw access to individual user data. Yet today's platforms, web browsers and mobile operating systems alike, still expose primarily individual-level data-access APIs, largely inherited from the POSIX read/write abstraction, and then attempt to constrain misuse through permissions and user prompts. This API mismatch drives pervasive tracking and backend data collection, entrenching a long-running privacy/utility arms race that treats privacy as a tax on utility. The result is mounting friction, complexity, and mistrust among the public, platform providers, and application developers – a dynamic set to only intensify in the AI era.

This article examines the W3C Attribution API, a draft browser standard for privacy-preserving advertising measurement, as a concrete and timely attempt to break this cycle. The Attribution API enables population-level conversion attribution – a core function of online advertising – without cross-site tracking or backend personal-data collection, using encrypted reporting and differentially private aggregation with on-device privacy enforcement. The authors of this paper, together with several collaborators, developed the on-device privacy architecture of this draft standard, grounding it in a strong, mathematically rigorous privacy guarantee for individual users that remains practically deployable.

We present the authors' own perspective on Attribution, focusing on its significant but as-yet unrealized potential to qualitatively improve web privacy, as well as on the major technical, legal, and economic roadblocks to widespread adoption. We argue that for Attribution to truly supplant tracking-based approaches, it must satisfy three criteria: (1) *high utility*, with robust and observable behavior even under adversarial conditions; (2) *rigorous privacy*, enabling platforms to incentivize adoption while tightening raw-data access; and (3) *broad accessibility*, so that both large and small ad-techs and advertisers can use the API cheaply and effectively. Numerous obstacles remain before these conditions can be met. We map these roadblocks and call on the research community to help advance Attribution so it can fulfill its promise as a viable, privacy-preserving alternative to tracking on the web, while briefly noting how similar benefits could extend beyond the web if aggregation were elevated to a first-order, operating-system-managed primitive – with the potential to de-escalate the privacy/utility arms race and help rebuild trust among users, platforms, and developers.

1 Introduction

Despite major progress in privacy technologies and platform controls, the dominant model for data-driven functionality feels stuck – on the web, in mobile operating systems, and beyond. Platforms and users increasingly seek to restrict access to sensitive user data, while applications rely on that same data to deliver core utility, including personalization, analytics, model learning, and now generative and agentic AI. The result is a persistent arms race. Platforms respond with tighter controls, such as disabling third-party cookies in some browsers and introducing increasingly granular permission systems for sensitive signals like location. Applications, in turn, adapt through varied tactics to recover data access paths: repeated or strategically timed permission prompts [32], mobile background components [22, 28, 31, 34, 41], device or browser fingerprinting, and other techniques that enable continued collection and backend aggregation

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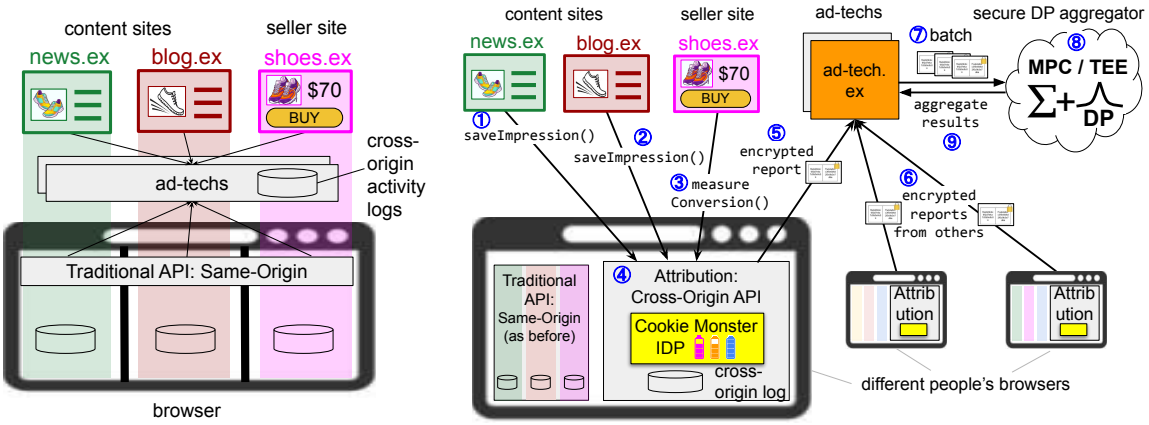

(a) **API/workload mismatch.**

(b) **New W3C Attribution API with our IDP architecture.**

Fig. 1. **The current API/workload mismatch in web advertising and how platform-managed aggregation changes it.**

beyond what is strictly required for primary functionality. In practice, privacy is treated as a tax on utility, and the ecosystem pays the price in friction, complexity, and growing mistrust.

We argue that a key technical driver of this dynamic is an *architectural mismatch between modern, aggregation-oriented workloads and the raw-data access interfaces* that platforms expose. Even when applications ultimately need population-level trends or learned models, platforms primarily offer POSIX read/write-style access to raw user data, and then attempt to constrain downstream use through permissions and user prompts. This is structurally misaligned: permissions and prompts try to restrict behavior that the data-access API itself encourages. In the absence of a high-utility, platform-managed path for aggregation, developers fall back on the *tracking pattern*: collecting individual-level data through ad hoc mechanisms, retaining it in backends, and aggregating later. This approach amplifies privacy risk through pervasive tracking and entrenches brittle, collection-heavy designs that are difficult for platforms to govern and for users to trust.

Web advertising provides a concrete and influential example of this mismatch (Fig.1a). Browsers are designed around isolation across origins (top-level domains), stemming from the same-origin policy. Advertising measurement, however, is inherently a cross-origin workload: it requires linking ad impressions on content sites (e.g., *news.ex*, *blog.ex*) to conversions on seller sites (advertisers, e.g., *shoes.ex*). In the absence of a structured browser API for this task, advertisers and ad-techs have long relied on workarounds such as third-party cookies, fingerprinting, and backend data exchanges to perform *conversion attribution measurements* – a foundational, aggregation-oriented function for optimizing advertising. These mechanisms undermine the intent of the same-origin policy, yet they persist largely because browsers historically offered no high-utility, privacy-preserving alternative.

This situation is now changing with an important new standard taking shape in the W3C: *Attribution* (https://www.w3.org/TR/attribution/), a browser API designed to support conversion attribution measurement without cross-site tracking or backend raw-data collection. Fig.1b shows its architecture. Content sites register ad impressions with the browser; seller sites later request encrypted reports indicating which ads a user previously saw; and only differentially private aggregate results across many users are released via secure multiparty computation (MPC) or trusted execution environments (TEEs). We, the authors of this paper, along with our collaborators, developed the API's differential

privacy architecture, establishing the formal definitions, analysis framework, and on-device enforcement mechanisms – known as Cookie Monster [38] – that ground Attribution in individual differential privacy (IDP) [12, 14, 18]. Under this architecture, each browser locally enforces per-epoch privacy budgets and deducts privacy loss before releasing any encrypted report; if an epoch's budget is insufficient for a requested report, the browser nullifies the contribution of that epoch's impression data in the encrypted report. Intuitively, this makes Attribution a *cross-origin interface with single-origin–like privacy*: information about a user's activity across sites is formally hidden behind a browser-controlled, *device-epoch–level IDP guarantee*. Attribution with our IDP-based core is now undergoing standardization with participation from all major browser vendors and is expected to enter public review in 2026.

This paper presents its authors' own perspective on the Attribution API: the promise we believe it holds to move web advertising off today's tracking-based default; the steep roadblocks that currently stand in the way of realizing this promise; and the broader significance of Attribution as the first concrete instance of a more generic, wider-reaching, platform-managed aggregation API. Our central claim is that Attribution represents a rare opportunity to qualitatively improve privacy on the web – not by further constraining tracking, but by offering a viable, high-utility alternative that could allow tracking-based approaches to be deprecated altogether.

However, whether Attribution can meaningfully shift the default away from tracking-based measurement depends on overcoming a set of significant roadblocks – technical, legal, economic, and organizational – that remain poorly understood and are therefore tempting to dismiss as fatal based on intuitive arguments, as some already have for related efforts [3]. Drawing on our experience designing the API's privacy architecture and engaging in its standardization to address additional technical challenges [37], we argue that many of these roadblocks are not insurmountable. Instead, they reflect deep technical problems that demand careful analysis and principled engineering. The goal of this paper is to identify these roadblocks, explain their technical origins, and outline research directions that the computing community should pursue to help Attribution move online advertising away from persistent tracking and toward structured, platform-controlled interfaces.

While this article centers on web advertising, the lessons from Attribution point to a broader architectural opportunity. The same mismatch between aggregation-oriented workloads and raw-data access APIs recurs across platforms, including mobile operating systems, where sensitive signals such as location are routinely accessed not only for primary, user-facing functionality but also for secondary, aggregation-driven uses such as analytics, recommendation, advertising, and reselling to data brokers. Attribution offers an early glimpse of how elevating aggregation to a first-order, platform-managed primitive could help de-escalate today's privacy/utility arms race more broadly – by reducing incentives for pervasive collection and helping rebuild trust among users, platforms, and developers. We urge the research community to take on this broader challenge, building on the momentum that Attribution's standardization can create.

2 Background

While this paper does not rely on technical depth for Attribution, we provide here a brief description of the API and of the on-device privacy architecture we developed for it. The goal is to ground the discussion of promise and roadblocks, and to provide a basic entry point for readers who wish to understand our perspective of jointly enforcing privacy and utility, which we believe is key for adoption. Full technical details appear in our prior work [37, 38] and in the Attribution specification itself [39].

2.1 Attribution API

We begin with additional detail on the workflow in Fig.1b. It involves four principals interacting with the browser via the Attribution API: (1) *impression sites* (e.g., *news.ex*, *blog.ex*) register ads via `saveImpression()`; (2) *conversion sites* (e.g., *shoes.ex*) call `measureConversion()` to request encrypted reports of which ads a user saw; (3) *intermediaries* (e.g., *adtech.ex*) receive encrypted reports; and (4) *aggregation services* (e.g., divviup.org) aggregate reports and add DP noise using trusted MPC or TEE. We refer to sites that issue measurement requests (typically conversion sites or intermediaries) as *queriers*.

The browser controls information flows as follows: impressions are stored locally, conversions trigger encrypted reports, and aggregators release only DP-protected statistics. A typical sequence is: ①–② impression sites record ads; ③ a conversion site calls `measureConversion()`; ④ the browser fetches relevant impressions, checks local privacy budgets, and ⑤ generates encrypted reports; ⑥–⑨ reports are batched, aggregated, and released with DP noise.

Critically, *privacy enforcement happens on-device* ④. Before releasing a report, the browser deducts the corresponding privacy loss from a local *privacy budget store*, returning null reports when budget is insufficient. This on-device enforcement, which is designed both for utility and privacy uses the privacy architecture we developed as part of two papers: Cookie Monster [38], which first introduced the definitional framework, formal analysis, and basic privacy accounting algorithm that permits this on-device enforcement; and Big Bird [37], which extended Cookie Monster with a set of utility-protecting mechanisms that preserve Cookie Monster's privacy guarantees while preventing malicious attacks against utility that can arise in a platform AI such as Attribution that aims to serve many competing actors running private measurements on user cross-site data.

2.2 On-Device Privacy and Utility Architecture

Fig.2 zooms into step ④ of the Attribution workflow, which executes entirely on the user's device and constitutes the core of our contribution to the draft standard. At this point, the browser must decide which cross-site information may contribute to a measurement request, under strict privacy constraints, while simultaneously supporting many independent queriers competing for limited privacy budget over time. Indeed, Attribution is a *platform-level* API: a single browser instance serves many queriers issuing overlapping measurements over a user's cross-site activity. While enforcing a single global privacy budget would be sufficient to ensure privacy, it would leave the system vulnerable to *utility depletion attacks*, in which aggressive or malicious queriers exhaust shared budget and deny meaningful service to others.

To address this challenge, we designed an on-device privacy architecture that treats *privacy and utility as co-equal, first-class concerns*. At its core is Cookie Monster [38], which provides the formal privacy foundation for Attribution. Cookie Monster introduced the definitions, analysis framework, and accounting algorithms needed to reason about privacy loss in this highly non-standard setting,

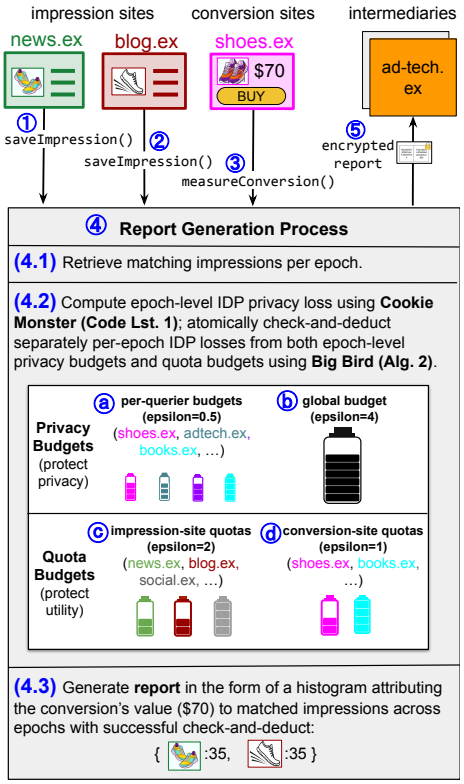

Fig. 2. **Detailed Attribution architecture: on-device privacy- and utility-related mechanisms.**

grounding Attribution in a previously-known but little-used form of DP, namely *individual differential privacy (IDP)* [12, 14, 18]. Under IDP, each browser enforces a device-epoch–level privacy guarantee, deducting privacy loss based on a user's *actual* contribution to a measurement rather than a population-wide worst-case bound.

Concretely, the browser partitions local data into non-overlapping epochs (e.g., weeks) and charges privacy loss only for epochs that contain impressions relevant to a given conversion; epochs with no relevant impressions incur zero privacy loss. Formally, this enforces a *device-epoch–level IDP guarantee*, enforced independently by each browser, while substantially optimizing privacy loss accounting compared to traditional DP formulations, which would typically incur privacy loss even for devices and epochs that contribute no relevant data. For workloads such as advertising – where users participate in measurements very unevenly because they see different ads and purchase different products – this IDP-based design both enables autonomous, browser-side control over user privacy loss (a key property for a consumer platform) and significantly improves utility by aligning accounting with actual user contributions.

Big Bird [37] extends the Cookie Monster privacy core with a utility-protective layer designed specifically for open, multi-querier platforms. While Cookie Monster ensures that privacy budgets are never exceeded, Big Bird introduces *quota budgets* that regulate how the global device-epoch budget may be consumed across queriers. These quotas bound not just the privacy loss caused by each separate querier: since queriers are often intermediary sites hidden from the users, they could replicate themselves into many different identities, exhausting the shared global budget via what would resemble a Sybil denial-of-service attack against the global budget. Rather, the quotas bound the contribution of distinct top-level sites that the user intentionally visits – notably impression sites and conversion sites – toward privacy loss consumption within the shared global privacy budget. This protects against depletion attacks that would otherwise permit one actor to crowd out others.

Crucially, Big Bird's quota enforcement preserves Cookie Monster's privacy guarantees: privacy loss accounting remains unchanged, while utility becomes resilient to adversarial or highly imbalanced workloads. Together, Cookie Monster and Big Bird form a unified on-device architecture that delivers strong, formally grounded privacy while ensuring that this privacy remains usable in a competitive platform setting.

2.3 Example Execution

To illustrate the combined privacy and utility behavior of our privacy architecture, we run through an example that based on the workflow from Fig.1b, now with concrete values of various parameters. The example uses a single attribution query corresponding to the evaluation of a creative developed by an ad-tech. A user sees a shoe ad on *news.ex* during epoch e_1 and another on *blog.ex* during epoch e_2. In epoch e_3, the user purchases \$75 worth of shoes on *shoes.ex*. The ad-tech *adtech.ex* issues a `measureConversion()` request to evaluate the performance of its creatives: how much revenue do the different creatives bring to the advertiser *shoes.ex*.

To process this request, the browser identifies impressions relevant to the conversion and evaluates whether sufficient privacy budget and quota capacity remain. Under Cookie Monster's IDP-based accounting, privacy loss is charged only for epochs that contain matching impressions (e_1 and e_2). Epochs with no relevant impressions incur zero privacy loss. For each relevant epoch, the charge is bounded by $(\text{value}/\text{maxValue}) \cdot \epsilon = (75/150) \cdot \epsilon$, using parameters supplied in the `measureConversion()` call (the system ensures that value $\leq$ maxValue). In contrast, with traditional (non-individual) DP, the privacy loss in all relevant epochs would have been ϵ, whether a relevant impression existed or the purchased shoe version cost less than the maximum value of \$150.

At the same time, Big Bird's quota checks ensure that *adtech.ex* cannot consume more than its fair share of the device's global privacy budget. Even if *adtech.ex* were to issue many such requests, quota limits would prevent it from exhausting shared budget and degrading utility for other queriers. If any quota or budget check fails, the browser simply excludes the affected epoch's data from the encrypted report, without deducting privacy loss.

The encrypted report is then returned to the ad-tech and aggregated with encrypted reports from other users who have purchased the shoes, using MPC or a TEE and with noise added before release. This execution illustrates the central design principle of the architecture: sparsity-optimized IDP accounting maximizes utility for typical workloads, while quota-based protections preserve that utility in the presence of competition or adversarial behavior – all without compromising privacy.

3 Attribution's Promise and Roadblocks

The background above describes the Attribution API and the on-device privacy and utility architecture that enables it. These mechanisms show that a core advertising measurement workload can, *in principle*, be supported without cross-site tracking or backend raw-data collection, while enforcing browser-controlled privacy guarantees. However, technical proof-of-concept alone is not enough. Whether Attribution can meaningfully shift web advertising away from its tracking-based default depends on more than the existence of a privacy-preserving API. In this section, we present the authors' perspective on Attribution's promise and its remaining roadblocks, and argue that for Attribution to realistically supplant tracking-based approaches, it must satisfy three technical conditions – high utility, rigorous privacy, and broad accessibility across the advertising ecosystem – and be supported by broader economic and legislative incentives that favor its adoption over tracking-based techniques.

3.1 The Promise: A Path Off Tracking

Attribution represents a rare and qualitatively different opportunity to improve privacy on the web. Unlike prior privacy interventions, which have largely focused on constraining or informing users about tracking, Attribution offers a plausible path to *replacing* tracking-based measurement altogether with a platform-managed alternative.

Historically, web privacy efforts have operated in an adversarial mode. Regulators and some browsers have sought to restrict tracking through mechanisms such as third-party cookie blocking [1], IP address obfuscation [2], and user-consent requirements [13], while advertisers and ad-techs have responded by developing new techniques to recover data access despite these roadblocks, such as remote fingerprinting that can recognize users across site visits without the need for third-party cookies. This dynamic has proven difficult to escape because it leaves the underlying substrate unchanged: raw, individual-level data remains the primary interface through which functionality is built. As long as this remains true, tracking-based designs retain a structural advantage over voluntary adoption by sites and intermediaries of more privacy-preserving alternatives, like federated learning.

Attribution challenges this default by having the platform provide a structured API that directly serves aggregation-oriented workloads. Rather than relying on users to police raw cross-site signals, the browser itself mediates measurement through a narrowly scoped interface that releases only differentially private aggregates. This shift is subtle but profound. It moves privacy enforcement into the platform, and it aligns the interface exposed to developers with the actual needs of (at least part of) the workload.

If successful, this shift could change incentives across the ecosystem. For advertisers and ad-techs, Attribution offers the possibility of obtaining essential measurement signals without incurring the legal, reputational, and operational challenges associated with pervasive tracking and backend data retention. For browsers and regulators, it creates a

credible basis for tightening raw-data access without unilaterally degrading functionality that, to a large extent, funds the open-web enterprise. For users, it promises a form of privacy that is not mediated primarily through prompts and settings, but is instead enforced by construction.

The promise of Attribution is therefore not merely incremental privacy improvement, but a *change in default*. Rather than asking how tracking can be constrained or tolerated, Attribution asks whether it can be rendered unnecessary. In doing so, it offers a path toward de-escalating a long-running privacy–utility arms race and replacing it with a more stable equilibrium based on structured, platform-managed aggregation. Although Attribution focuses on advertising on the web, success in this domain could establish an important reference point for other settings in which tracking is already dominant or poised to become so, including in mobile platforms, smart spaces, generative AI, and mandated web-search data releases. We discuss these broader implications in §5.

3.2 What Success Would Require

Attribution's promise hinges not merely on its ability to demonstrate that privacy-preserving advertising measurement is possible, but on whether it can serve as a *credible replacement* for tracking-based approaches in practice. In today's ecosystem, tracking is not used because it is elegant or principled, but because it delivers a combination of utility, flexibility, and accessibility that no alternative has yet matched. As long as this remains true, tracking-based designs will persist, even in the presence of privacy-preserving APIs.

We argue that for Attribution to realistically supplant tracking as the default mechanism for advertising measurement, it must satisfy three technical requirements. First, it must deliver ***high utility***: its outputs must be sufficiently accurate, robust, and usable to support ad-measurement workloads roughly on par with the current tracking-derived data. Second, it must provide ***rigorous privacy***: its guarantees must be strong enough to justify preferential treatment by browsers and regulators (such as enabling it by default with the possibility to opt-out), enabling access to the API without the same friction or scrutiny imposed on raw-data collection. Third, it must achieve ***broad accessibility***: the API must be usable not only by large incumbents with substantial data, infrastructure, and financial resources, but also by smaller advertisers, ad-techs, and newcomers that are essential for a competitive advertising ecosystem.

Failure to meet any one of these requirements risks relegating Attribution to a complement, rather than a replacement, for tracking, thereby forfeiting its potential to materially improve society's privacy posture. In the remainder of this section, we examine each requirement in turn, argue why there is reason for optimism at a technical level, and identify the major roadblocks that must be overcome for Attribution to succeed. §4 then outlines concrete research directions through which the computing community can help turn Attribution's promise into a durable, real-world alternative to tracking.

3.3 High Utility

The bar. The first and most immediate requirement for Attribution's success is high utility. In practical terms, this means that Attribution-based measurement must be *nearly as useful as measurement based on the tracking data it seeks to replace*. This is an ambitious bar, but we believe it is the correct one to aim at. Anything substantially weaker risks positioning Attribution as a secondary signal that will only complement, rather than displace, tracking-based workflows.

Why the bar may not be out of reach. At first glance, this bar may seem unrealistic. Attribution-based measurements inevitably introduce error through several mechanisms: noise added for DP; bias from value clipping (also present in traditional DP); and, under our IDP formulation, missing contributions when privacy budgets are exhausted and reports

are nullified. The latter is specific to IDP-based on-device enforcement, where the presence or absence of a report itself is privacy-sensitive because it depends on user data and local budget state, and therefore cannot be revealed to queriers.

However, this comparison is often made against an idealized notion of tracking-based conversion attribution "ground truth" that does not exist in practice. Tracking-derived conversion attribution measurements are themselves noisy and incomplete, due to cookie resets, default third-party cookie blocking in browsers such as Safari and Firefox (and with the users having this possibility in Chrome), IP address obfuscation and NATs, and other sources of imprecise attribution. As a result, the true utility gap between API-based measurement and tracking-based measurement remains largely unmeasured and poorly understood.

From this perspective, the relevant comparison is not between noisy aggregates and perfect conversion-attribution data, but between two imperfect measurement regimes with very different failure modes. Attribution reveals the ground truth noisily, while tracking attempts to reconstruct it incompletely. Determining which approach yields better utility for real advertising workloads is ultimately an empirical question. To our knowledge, no comprehensive, end-to-end evaluation has yet been conducted to quantify this tradeoff, and such measurements are urgently needed to ground debates about Attribution's viability.

Roadblocks. Even if near-parity utility may be achievable in principle, reaching it in practice faces significant roadblocks. One challenge is robustness under adversarial conditions. Advertising measurement has long contended with manipulation such as click fraud and data poisoning, but Attribution introduces new difficulties by operating over encrypted data and privacy-constrained aggregates. Preventing or detecting manipulation without access to raw data is fundamentally harder, and while recent work has begun to explore this problem [8], robust defenses that work well specifically for this API remain an open research problem. Our own work on budget depletion attacks highlights how adversarial behavior can target utility even when the data itself is not manipulated [37].

A second challenge lies in usability and observability. Practitioners today rely heavily on raw data not only for final measurements, but also for debugging, exploration, and iterative refinement. Attribution reports, by contrast, are privacy-limited resources: each report can be used only a small number of times, and errors or mis-specifications cannot be undone by re-querying historical data. Supporting effective debugging and interpretability in this setting will require new tools and workflows tailored to privacy-preserving measurement, rather than direct analogues of existing raw-data pipelines.

Finally, high utility depends on expressiveness. The current Attribution API supports a narrow class of pre-specified, histogram-based aggregations with attribution logic fixed in advance. This is sufficient for some reporting tasks, but far from the full range of optimization and machine-learning workflows that advertisers and ad-techs run today. Supporting richer aggregation logic, delayed query selection, and learning-based workloads without sacrificing privacy or security remains a major technical challenge.

Overall, we advance the opinion that high utility is neither guaranteed nor impossible for Attribution, but that achieving it will require sustained research effort across robustness, usability, and expressiveness, as well as careful empirical evaluation with appropriate baselines. Without such progress, Attribution risks falling short of the utility threshold needed to credibly replace tracking-based measurement.

3.4 Rigorous Privacy

The bar. The second requirement for Attribution's success is rigorous privacy. Taking a pragmatic rather than idealized perspective, we view strong privacy not only as an end in itself, but as a prerequisite for changing incentives. Even

a high-utility API is unlikely to replace tracking if it merely becomes an additional data source alongside raw-data collection. For Attribution to plausibly displace tracking as the default, it must provide privacy protections strong enough to justify preferential treatment by browsers and regulators – for example, enabling access to Attribution reports *by default*, without user permission prompts, while imposing greater friction or scrutiny on raw-data collection. In this sense, privacy is not only about protecting users, but also about creating the conditions under which tracking can be actively dis-incentivized. Without a strong, defensible – if not ideal – privacy semantic, Attribution risks being adopted only opportunistically, rather than serving as a foundation for a broader shift in platform policy and regulation.

Why our privacy semantic is a solid foundation. We designed Attribution's privacy architecture to provide a guarantee that is not only formally meaningful, but also well matched to the realities of a long-lived, consumer-platform-managed API. Attribution must operate continuously over time, mediate access to sensitive data across many independent and mutually distrustful queriers, and remain usable without centralizing user privacy control in any single backend aggregation service. These constraints limit which privacy semantics are viable in practice.

Our design centers on a *global device-epoch–level individual differential privacy (IDP)* guarantee, enforced independently by each browser. Two aspects of this semantic are essential.

First, the device-epoch privacy unit provides a well-defined notion of data protection over time. This stands in contrast to user-level DP with automatic per-epoch budget refresh, a pattern commonly used in industry [4, 33, 35] and often conflated with the device-epoch formulation. Resetting a user-level budget does not bound cumulative privacy loss and is better understood as a rate limiter on an otherwise unbounded user-level privacy metric, which on its own has no formal privacy interpretation. By contrast, the device-epoch formulation bounds the privacy loss associated with an individual's data *within each epoch*, yielding a well-defined DP guarantee over that epoch's data. This guarantee can be interpreted as a formal *data-protection* guarantee, while still permitting a rate-limited interpretation of unbounded user-level privacy loss over long horizons. Intuitively, it ensures that what a user did during a particular epoch (e.g., a week) remains protected even if arbitrarily many queriers later attempt to extract information about that same epoch. This distinction is critical for Attribution, which is explicitly designed to operate continuously as data and queries arrive indefinitely: in such settings, a data-protection guarantee is the only well-defined privacy guarantee we know how to provide without utility degrading to zero.

Second, adopting individual differential privacy aligns privacy enforcement with both the workload and the platform trust model. IDP measures privacy loss with respect to a user's *actual* contribution, allowing each browser to enforce privacy autonomously based on locally observed data. As discussed in §2, this enables efficient accounting that exploits the inherent sparsity of conversion-attribution workloads, yielding substantially higher utility than traditional DP formulations. Equally important, IDP keeps the browser – rather than any backend service – at the center of privacy control. While aggregation services (e.g., MPC- or TEE-based) must be trusted to securely and noisily aggregate encrypted reports, they remain stateless with respect to user privacy management. This separation allows Attribution to support multiple trusted aggregation services simultaneously, without coupling privacy guarantees to any single backend or execution environment, and enables a competitive ecosystem of aggregation services and queriers around the platform API.

These design choices yield a privacy semantic that is strong in a precise technical sense and operationally compelling for the web platform. They support continuous operation, decentralized enforcement, and multi-aggregator flexibility, while providing a bounded and defensible privacy guarantee that can plausibly serve as a foundation for platform policy and regulatory decisions.

Roadblock: choosing parameters for a strong semantic. Even with a semantic that is strong *in principle*, a central challenge remains: what constitutes "strong privacy" *in practice*? For Attribution's device-epoch–level IDP guarantee, this question reduces to the choice of deployment parameters – most notably the privacy budget ϵ and the epoch length – both of which have profound implications for utility.

From a theoretical perspective, values of $\epsilon \leq 0.1$ offer strong privacy protection. From a practical perspective, extracting meaningful utility from DP aggregations in most deployed systems typically requires values orders of magnitude larger (e.g., the U.S. Census' trusted, one-time release enforced $\epsilon \approx 7$). With Attribution's extremely ambitious goal of supporting continuous, global privacy enforcement across many competing queriers – and while contending with malicious behavior such as budget depletion attacks – the effective per-epoch global privacy budget may need to be substantially higher to achieve good utility – potentially in the tens or hundreds. At such scales, traditional theoretical interpretations of ϵ become impossible to apply. Compounding this challenge, the DP literature offers little guidance on how to select epoch lengths (e.g., days, weeks?) in a principled way for long-lived systems. Yet, both of these parameters interact in complex ways when it comes to utility and privacy protection, e.g., shorter epoch lengths may permit lower global budgets for the same utility, but how does this choice compare to others in privacy protection?

What it will take to overcome this roadblock. We argue that for Attribution to have a realistic chance of achieving both high utility and rigorous privacy, we will need to combine *(1) pragmatism in parameter selection* with *(2) uncompromising commitment to a strong semantic.*

(1) Pragmatism in parameter selection: Empirical studies of differentially private machine learning suggest that, against known practical attacks, observed privacy leakage can sometimes be substantially lower than worst-case theoretical bounds would imply [17, 23]. At the same time, empirical protections can be undermined as stronger attacks are developed [7]. While it remains unclear how these observations translate to advertising measurement workloads, they motivate a hybrid evaluation approach: assessing concrete instantiations of a strong privacy semantic not only through worst-case theoretical analysis, but also through systematic and continuously maintained attack suites and stress testing that illuminate practical privacy–utility tradeoffs across parameter choices such as privacy budgets and epoch lengths.

(2) Uncompromising commitment to a strong semantic: While parameter choices may be pragmatic, the underlying privacy semantic should not be weakened. A well-defined guarantee such as global device-epoch–level IDP holds regardless of attack strategy. If stronger attacks emerge, the response is straightforward: extend the attack suite, re-evaluate parameters, and update deployments. By contrast, weaker or ad hoc approaches—such as anonymization schemes tuned to a fixed set of attacks or weakened DP notions—often lack clear failure modes. Addressing new attacks in such systems may require redesigning protection mechanisms or even revising standards, creating costly disruptions for queriers and platforms.

Overall, overcoming the roadblock of defining what constitutes "strong privacy" requires both technical advancements and a change in perspective. First, the research community must develop principled, attack-driven methodologies for configuring strong privacy semantics in real systems. Second, we must acknowledge the reality that with the current privacy theory, large-scale DP deployments will inevitably need to combine strong theoretical guarantees with empirically grounded parameter choices. Without this shift in perspective, Attribution – and similar systems – risk being judged against unattainable ideals rather than evaluated as serious candidates for replacing today's tracking-based defaults.

3.5 Broad Accessibility

The third requirement for Attribution's success is broad accessibility. For Attribution to function as a viable replacement for tracking – and to succeed as a W3C standard – it must be usable not only by large incumbents, but also by small advertisers, ad-techs, and new entrants. An API that is technically sound but economically or utility-wise inaccessible to small players risks entrenching existing power asymmetries rather than improving the web ecosystem. Two concerns stand out.

Roadblock: Cost and scalability. Attribution relies on privacy-preserving aggregation via MPC-based services, which are significantly more expensive than traditional raw-data pipelines. While such costs may be tolerable for large advertisers and major ad-techs operating at scale, they pose a serious barrier for smaller actors. If aggregation costs scale poorly or require substantial fixed overhead, small advertisers may be priced out of meaningful participation. This raises a fundamental tension: the very mechanisms that enable strong privacy may also concentrate access among the largest players unless cost-effective architectures and shared infrastructure models are developed.

Roadblock: Utility for small advertisers. DP is inherently less accurate on small datasets. Small advertisers and niche campaigns naturally generate fewer impressions and conversions, resulting in sparse reports where DP noise and clipping dominate the signal. As a result, these actors may be effectively *noised out* of the API, unable to obtain actionable insights even if they can afford access. This problem is not merely a matter of tuning parameters: it reflects a deeper mismatch between privacy-preserving aggregation and low-volume measurement workloads.

Viewing roadblocks through an engineering lens. These concerns have been raised repeatedly in prior discussions of privacy-preserving measurement systems [3]. Rather than treating them as fatal, we argue that they should be approached as open technical and engineering challenges.

For example, ongoing work in cryptography explores *asymmetric MPC* protocols [5], in which a large helper can shoulder a disproportionate share of the MPC computation without requiring additional trust compared to smaller helpers. If such approaches mature, they could allow secure aggregation services to benefit from the economies of scale of large providers, while preserving the same trust assumptions for browsers and users. Until then, TEE-based aggregation offers a plausible – though imperfect from a threat modeling perspective – alternative that may reduce costs enough to make Attribution accessible to smaller players.

A second example concerns the utility gap faced by small advertisers. In §4.2, we propose reframing the problem by abandoning the assumption that all queriers must be served independently. Instead, small queriers could be paired with other, similar queriers – small or large – to perform joint, cross-querier measurements over shared encrypted-report pools. Because reports are encrypted and privacy is enforced by each browser at a global, all-querier level, encrypted reports can be shared across queriers for joint aggregation without introducing additional privacy loss for users, provided that no auxiliary information beyond the encrypted reports is exchanged. In this model, small advertisers may extract meaningful utility through collective measurement, even if they lack sufficient data to justify dedicated measurements on their own. This perspective offers a constructive way to navigate a very fundamental privacy–utility tradeoff at very small scale, rather than treating it as a fatal limitation of the API.

3.6 Fundamental Limits

The preceding sections described significant roadblocks to Attribution achieving its promise of shifting the web away from tracking. While these challenges are substantial, we believe many can be addressed through a combination of

technical advances and changes in perspective. In §4, we outline research directions that may help make progress on these fronts. That said, it is equally important to acknowledge a fundamental limit of what Attribution can achieve.

Attribution will not eliminate all forms of cross-site tracking. Certain advertising functions – most notably targeting – inherently rely on access to individual-level data and cannot be expressed as aggregation across many users. Our vision is therefore not that Attribution replaces all tracking-based functionality, but that by supporting a large portion of the advertising workload – especially non-targeting measurement and optimization – via a strong-privacy, platform-managed API, it enables a meaningful rebalancing of incentives.

If aggregate measurement tasks can be supported at high utility under strong privacy guarantees, platforms and regulators may gain the latitude to dis-incentivize tracking more aggressively than is feasible today. For example, aggregate measurement could be enabled by default, while access to individual-level data could require explicit user consent or face stricter enforcement against circumvention techniques such as fingerprinting. In this sense, Attribution's privacy guarantees are not meant to stand alone, but to serve as an enabling foundation for broader shifts in platform policy and regulatory enforcement – shifts that are currently constrained by the advertising ecosystem's deep reliance on tracking.

4 Research Challenges

The preceding section argued that Attribution's success hinges on three properties – high utility, rigorous privacy, and broad accessibility – and identified roadblocks that make each difficult to achieve in practice. This section distills those roadblocks into a non-exhaustive set of research challenges that we urge the computer research community to take on promptly and constructively, so that Attribution has a realistic chance of supplanting tracking rather than merely coexisting with it as an additional signal, delivering utility gains with no privacy benefits.

This is also the right moment for such work. Attribution is no longer a speculative design, yet not yet ossified into an immutable standard and deployed web protocol. Decisions being made now – about privacy semantics, parameter regimes, interfaces, supporting infrastructure, and interaction with regulatory frameworks – will largely determine whether privacy-preserving aggregation becomes a first-class alternative to tracking, or settles into a marginal role with limited or no impact. Progress on the challenges below can shape not only the future of Attribution itself, but the broader viability of deploying strong-privacy mechanisms in large-scale, competitive, complex, real-world ecosystems. We see researchers from computer systems, DP theory, cryptography, machine learning, law, and economics having great potential to contribute to Attribution.

4.1 Develop Querier Tools for Debugging, Bias, and Quality Control

Problem statement. A central roadblock to high utility (§3.3) is not only statistical accuracy in the narrow DP sense, but *operational utility*: queriers must be able to determine whether a measurement is functioning, interpret anomalous results, and iterate on parameters (e.g., value, maxValue, and ϵ in `measureConversion()`, §2.3), attribution logic, and batch sizing – all without access to raw data.

In tracking-based pipelines these tasks are routine: queriers inspect logs, replay computations, and run post-hoc analyses over retained datasets. Attribution removes these capabilities by design. Reports are end-to-end encrypted, clipped, and effectively single-use; queriers observe only noisy aggregates. A single misconfiguration can silently produce unusable or biased outputs, and diagnosing the issue may require collecting *new* reports and waiting for additional conversions.

This opacity is not unique to Attribution; it reflects a persistent gap in the DP literature. Most DP mechanisms are evaluated under correct use, focusing on asymptotic privacy–utility tradeoffs, while the systems question of how practitioners *debug and control quality* under privacy-constrained outputs is often treated as an implementation detail. For Attribution – and for any large-scale DP deployment intended to replace raw-data pipelines – that stance is untenable. If queriers cannot detect misconfiguration, degradation, or bias, the API will be perceived as brittle and will fail the high-utility bar regardless of its formal guarantees. Addressing this gap requires research, not just engineering: new DP mechanisms for iterative diagnosis under tight privacy budgets, and system architectures that expose actionable signals without reintroducing raw-data observability.

Importantly, the solution goes beyond simply releasing synthetic samples, which are often proposed as a remedy for opacity. Instead, the DP interface itself must expose debugging signals alongside query results – signals that help developers assess whether errors may stem from data quality, parameter choices, or query design. We do not yet know what such signals should look like. Some early ideas appear in debugging tools proposed for related systems [30]. Below we illustrate the challenge through one concrete example from our own work: measuring the bias introduced by IDP accounting. The goal is not to suggest that this is the only necessary tool, but to demonstrate that developing such tools is itself a research problem, requiring new theoretical foundations before practical implementations are even possible.

An example. Consider bias induced by IDP accounting. Under IDP, privacy loss is charged based on a user's *actual* contribution to a query, so users and epochs deplete budgets at different rates. Because budget state depends on user data, depletion events cannot be revealed directly; the browser must silently nullify out-of-budget contributions.

Over time, this creates *selection bias*: results become selectively missing in ways correlated with user behavior and workload. From the querier's perspective, aggregates drift, but the cause is invisible. This is not an edge case but a structural consequence of contribution-sensitive accounting combined with hidden budget state.

Attribution's design provides a partial foothold: batch size is observable. This makes it possible, in principle, to estimate bias indirectly by privately measuring how many records in a batch still have sufficient budget to contribute. In many cases, queriers do not require continuous bias estimates; they only need to know when bias exceeds a threshold. This suggests sparse-vector-style monitoring mechanisms (e.g., SVT [19, 24] or Target-Charging Technique (TCT) [9]), where privacy cost is incurred primarily when thresholds are crossed.

Research challenges for the example. However, existing SVT/TCT theory does not yet support deployment in systems like Attribution. Prior analyses typically do not address (i) *IDP semantics*, where privacy loss should depend on whether a given user or epoch actually triggers a threshold event; (ii) *adaptive budget management* under explicit privacy filters in long-lived systems; or (iii) composition-efficient definitions such as Rényi DP or f-DP that are often necessary at real-world scale. Thus "use SVT/TCT" is not an engineering choice but an open DP-theory problem.

Even with a sound monitoring primitive, architectural questions remain: how to scope and store monitoring state (per querier? per measurement? tied to aggregation services?); how to treat epochs deemed too biased when queries target non-i.i.d. subpopulations; and how monitoring composes with utility-resilience mechanisms such as Big Bird's quota defenses. These design decisions determine whether monitoring remains lightweight, abuse-resistant, and compatible with on-device enforcement.

Querier-facing tools for debugging, bias detection, and quality control are therefore not conveniences but prerequisites for Attribution to meet the utility bar required to displace tracking. Addressing them demands advances in both DP theory and DP systems architecture. Although we illustrate the challenge through Attribution, similar issues arise

in any real DP deployment, suggesting that solutions developed here could influence the broader design of usable, high-utility privacy-preserving platforms.

4.2 Develop Methods for Learning and Small-Advertiser Workloads

Problem statement. A central concern – from both a utility and accessibility perspective (§3.3, §3.5) – is that Attribution's current design may disadvantage small advertisers and learning-based workloads. The API today supports single-advertiser, histogram-style DP aggregation, while differential privacy inherently degrades utility at small scale. Low-volume advertisers risk being effectively *noised out*, whereas large actors amortize noise across substantial data. At the same time, modern advertising depends not only on reporting but on continuous optimization and machine learning, which the current interface does not natively support.

These limitations are real at present, but they may not be fundamental. The research challenge is to determine how Attribution can be extended or restructured so that optimization and small-scale measurement become viable under strong privacy constraints rather than structurally excluded by them.

Research challenges. The first question is architectural: *what forms of learning are genuinely compatible with Attribution's structure?* Attribution is not standard federated learning (FL). It combines user-event-driven, on-device attribution and IDP accounting with MPC/TEE-based aggregation over encrypted reports. Conventional FL patterns – iterative model dissemination and gradient exchange – may not map cleanly onto this workflow. The community should therefore investigate learning paradigms that operate directly over noisy cross-device aggregates rather than per-device updates.

Learning from Label Proportions (LLP) [6, 21] is a natural candidate, since it trains from aggregate labels over *bins of examples* rather than per-example labels, aligning with Attribution's aggregation model. Yet LLP remains understudied, particularly under differential privacy. Key questions include: How does LLP compare, empirically and theoretically, to alternatives such as DP-SGD (often prohibitively costly in this architecture) or local-DP approaches (typically lower utility)? How does training quality degrade as advertiser scale shrinks? What algorithmic modifications are required for aggregate-based learning to function robustly under heterogeneous advertiser sizes?

A second line of research concerns *explicit support for small-scale regimes*. The current API assumes independent per-advertiser aggregation, but this assumption need not be fixed. One promising direction is native support for cross-advertiser queries: pooling encrypted reports across multiple advertisers, which is sound to do from a privacy perspective under Attribution's global IDP enforcement. Such pooling can increase effective sample sizes for low-volume advertisers, which could enable collective measurement and joint optimization tasks – such as conversion-rate learning – while preserving the privacy semantic. Open technical questions include how to structure grouping or clustering of advertisers to maximize signal for small participants; how to allocate noise across pooled queries; and how to design objectives that mitigate scale-driven performance disparities without weakening privacy guarantees.

The overarching question is whether aggregation APIs can be engineered so that small advertisers are not forced into isolation, but can participate in collective measurement and optimization under the same global privacy guarantees. Demonstrating viable technical pathways here would ground the broader debate about Attribution's competitive impact – a debate that warrants rigorous economic study (which we briefly discuss in §4.8) but should not be resolved by assuming that the technical design of Attribution is now set in stone.

4.3 Develop Methodology for Empirical Tuning of Privacy Parameters

Problem statement. As argued in §3.4, Attribution's privacy semantic – global device-epoch–level IDP – is strong and well-defined in principle. Yet its practical impact depends critically on deployment parameters, most notably the privacy budget ϵ and the epoch length. These parameters jointly determine both utility and privacy protection, and their selection currently lacks a principled methodology.

There is an unavoidable tension here. For a long-lived, globally enforced platform such as Attribution, theoretically conservative parameter regimes (e.g., small ϵ) are unlikely to sustain acceptable utility – let alone high enough to compete with tracking – across many queriers and adversarial workloads. Conversely, parameter regimes that yield high utility may fall outside traditional theoretical comfort zones. In practice, it is unrealistic to expect that a globally deployed, continuously operating IDP system will succeed under parameters chosen solely from worst-case theoretical bounds. The question is not whether pragmatism will be required, but how to make it *principled*.

Research challenges: principled empirical methodology. We believe that the research community should spearhead rigorous, attack-driven methodologies for configuring privacy parameters in large-scale, continuously operating DP systems. This requires moving beyond purely asymptotic interpretations of ϵ toward evaluation frameworks grounded in realistic, applicable threats. Concretely, researchers should design standardized attack suites – including membership inference, reconstruction, linkage, and adaptive querying attacks – tailored to Attribution-like workloads. Parameter choices should then be stress-tested against these suites across a range of privacy budgets and epoch lengths, quantifying empirical risk alongside utility.

Some empirical privacy suites already exist [20, 26, 29], but how to use them to guide deployment parameters for any given DP system is very much an open problem. What types of data would be needed from the DP system to make a rigorous decision? How do different parameters of a deployment, such as epoch length and ϵ impact privacy? Could Attribution natively support continuous measurement of vulnerability under attack suites, or must the methodology rely on periodic releases of datasets by participants and simulations of realistic workloads? The former approach would be compelling, if it were possible under the same IDP guarantees.

Moreover, parameter selection is not a one-time decision. Privacy risk evolves as attack techniques improve. A viable privacy-preserving platform must therefore incorporate a process for continual re-evaluation. This creates a governance challenge: establishing institutional mechanisms to monitor empirical privacy risk over time – and to adapt to it. This could include maintaining up-to-date attack suites, curating representative datasets for privacy evaluation, and establishing processes such as bug bounties or an auditing body that will periodically reassess parameter regimes in light of new attack capabilities. Browsers or independent bodies may need to assume responsibility for coordinating such evaluations, ensuring transparency, and updating deployments when risk profiles change.

Without such principled, attack-driven, and continuously updated methodology, debates over "strong enough" privacy will remain polarized and unproductive. With it, Attribution and other DP systems can be evaluated as serious engineering artifacts: systems whose parameters are chosen transparently, stress-tested empirically, and revised responsibly as the threat landscape evolves.

4.4 Develop an Accountability and Abuse Detection Layer for Attribution

Attribution's privacy protections can also obscure misuse. Because reports are encrypted and privacy enforcement occurs on-device, neither external overseers nor even platforms necessarily have direct visibility into whether the API is being exercised in ways consistent with its intended goals. For example, it may be difficult to detect whether a small

number of queriers are consuming disproportionate shares of global privacy budget, whether patterns consistent with budget-depletion attacks are emerging, or whether ecosystem participation is concentrating among a few dominant actors.

The research challenge is to design an *accountability layer* that provides aggregate, privacy-preserving visibility into system-level behavior without reintroducing individual-level observability. Concretely, this could include mechanisms for releasing differentially private summaries of budget utilization across queriers, concentration metrics over time, anomaly-detection signals indicative of coordinated depletion attempts, or transparency reports about aggregation-service load and query distribution. Such signals would need to compose cleanly with on-device IDP accounting, avoid revealing sensitive user data or querier strategy, and be robust to strategic manipulation.

This accountability problem is distinct from querier-facing debugging and quality control (§4.1): it is about providing aggregate visibility into *system behavior* for platforms, users, and regulators, without weakening the privacy semantic that makes Attribution a credible alternative to tracking.

4.5 Study Compatibility of Attribution with Privacy Regulations

Attribution's long-term impact will depend not only on its technical merits, but also on its treatment under privacy law and regulatory practice. As discussed in §3.4, a privacy-preserving API cannot displace tracking if it is regulated in the same way as raw-data collection. For Attribution to meaningfully shift incentives, it may need to receive differentiated treatment – for example, being accessible by default or subject to lower friction than tracking-based mechanisms that rely on third-party cookies, fingerprinting, or extensive raw-data retention.

Assessing whether current legal frameworks permit such differentiation is a question that is too important to leave open and should be addressed together with privacy experts. Researchers operating at the intersection of privacy technology and law should examine whether existing statutes and regulatory interpretations – across U.S. jurisdictions and internationally – can accommodate "preferential treatment" for privacy-preserving aggregation mechanisms grounded in strong, well-defined semantics such as device-epoch–level IDP, and moreover tuned based upon empirical privacy attack. If not, they should analyze what technical or architectural modifications to Attribution might bring it within such a category.

Key questions include: Can DP-based aggregation be interpreted as sufficiently minimizing personal data processing to justify lighter consent requirements? Under what conditions might regulators view IDP-enforced aggregation as materially distinct from cross-site tracking? What residual obligations would browsers or aggregation services retain under existing data protection frameworks? And what design choices – parameter regimes, enforcement transparency, auditability, user controls – would strengthen the legal defensibility of Attribution as a privacy-enhancing technology?

A related and potentially deeper question concerns relative utility. If Attribution were shown to deliver utility comparable to tracking-based approaches, could certain legal frameworks interpret continued reliance on tracking as disproportionate or excessive data collection? We do not know whether such utility parity is achievable – nor how best to evaluate it, which itself may require careful study. Nevertheless, it is important to analyze the full spectrum of potential regulatory differentiation: from lighter consent treatment for privacy-preserving aggregation, to more aggressive scrutiny of tracking once viable alternatives exist. Systematic, cross-jurisdictional analysis of these possibilities is essential if privacy-preserving architectures are to move from theoretical constructs to legally and economically viable replacements for tracking.

We would like to specifically encourage researchers to consider parallel standardization developments and how they may impact Attribution's prospects. The W3C Global Privacy Control (GPC) [16, 40], a draft standard presently being developed in parallel to Attribution, propagates user opt-outs for the sale or sharing of personal data under statutes that recognize such signals, including California's CCPA, Connecticut's CTDPA, New Jersey's NJPA, and analogous laws taking effect across additional U.S. states. Where GPC is legally recognized, businesses relying on tracking for measurement will face statutory constraints on processing user data for secondary purposes absent explicit consent – constraints that Attribution, by design, does not implicate in the same way. This creates a potentially meaningful complementarity: Attribution addresses the technical workload that drives measurement-oriented tracking, while GPC and the statutes it operationalizes apply legal pressure to the data-brokerage and cross-context sharing workloads that tracking also serves.

For researchers studying Attribution's regulatory fit, this interplay raises concrete questions: Does routing measurement through Attribution bear on data-minimization or necessity requirements under these statutes? As GPC recognition expands, does the asymmetry in regulatory cost between tracking-based and Attribution-based measurement grow in ways that tip adoption incentives? And how should Attribution's architecture evolve to remain clearly distinguishable from tracking under the frameworks that GPC reinforces? These questions underscore that Attribution's development should not proceed in isolation – it should engage with the broader legislative ecosystem in which Attribution will actually operate.

Importantly, these should not be treated as merely compliance exercises, but rather as *system design questions*. If Attribution is to serve as a viable alternative to tracking, its architecture must align with legal categories in ways that allow platforms and regulators to actively promote it relative to more privacy-invasive approaches. The standardization window is still open, and design decisions made now may determine whether Attribution can plausibly receive such differentiated treatment in practice.

4.6 Increase Scalability and Cost Efficiency of Attribution MPC

Attribution's aggregation layer relies on privacy-preserving computation, most prominently MPC-based secure aggregation services, largely based on Prio [10] or rather an enhanced version of it that is currently under IETF standardization [15]. While we lack deep expertise in MPC protocol engineering, we view scalability and cost efficiency of this component as a critically important research problem. If secure aggregation remains expensive or operationally complex, Attribution risks remaining a limited rollout – a niche exception to tracking rather than a viable replacement.

TEE-based aggregation may offer a pragmatic path to lower cost and operational simplicity. However, from a security perspective, it would be preferable to pair Attribution's differential privacy guarantees with stronger, cryptographic security assumptions. Achieving MPC instantiations that are economically competitive with traditional raw-data pipelines is therefore a major open systems and cryptography challenge.

Until aggregation can be made both scalable and cost-effective, two consequences follow. First, large-scale deployment will be difficult to justify relative to tracking-based infrastructure that is already optimized for scale. Second, smaller advertisers and ad-techs may be priced out by the overhead of secure aggregation, exacerbating the accessibility concerns discussed in §3.5. In other words, cryptographic strength alone is insufficient; it must be paired with economic feasibility.

There is a vast body of work on scaling MPC and reducing its cost, and we do not attempt to survey it here. Our goal is simply to underscore the importance of this line of work for Attribution's success and to offer a few suggestions to researchers in this space based on our experience with the system. In particular, MPC protocol designers should

consider the specific structure of the Attribution API, as well as the market forces and economies of scale under which such protocols would be deployed. Accounting for these constraints may yield designs that are cheaper, more scalable, and easier to deploy than protocols optimized for more abstract deployment models. We observed exactly this dynamic while developing Attribution's differential privacy architecture, and it is plausible that similar lessons apply to MPC protocol design.

One promising direction may be asymmetric MPC [5], in which a large participant shoulders most of the computational burden while smaller participants retain meaningful verification or checking roles under comparable trust assumptions. Such designs may enable economies of scale without collapsing the trust model into a single dominant aggregator. Exploring these architectures – and evaluating their concrete performance in Attribution-like workloads – is therefore an important research direction.

Finally, scalability is not purely a technical question; it is also a market question. A competitive ecosystem of aggregation providers is unlikely to emerge unless browsers demonstrate credible, long-term commitment to MPC-based aggregation. Without such commitment, privacy-preserving infrastructure risks becoming a temporary experiment rather than a stable platform primitive. Research on scalable MPC must therefore proceed alongside clear signals from the platform vendors themselves that enable a durable market for secure aggregation services. We urge platform vendors to establish such signals so a market may start to form.

4.7 Develop Fraud Detection and Prevention Mechanisms

Fraud mitigation is fundamentally complicated by Attribution's core design principles: reports are encrypted, secret-shared, and revealed only in aggregate. While this structure protects user privacy, it also makes identifying fraudulent reporters or impression manipulation extremely challenging, since individual-level signals are deliberately obscured.

As with MPC protocols more broadly, there is substantial ongoing research on certifying or validating inputs to secure aggregation (see [8] for a short list). Our goal here is not to survey this literature, but rather to encourage researchers to consider the specific setting of Attribution, which introduces distinct threat models, constraints, and opportunities that may influence protocol design.

One example we find particularly promising is recent work by Chairattana-Apirom, Tessaro, and Tyagi [8], which proposes Secret Share Attestation (SSA), a cryptographic mechanism tailored specifically to the Attribution (formerly PPA) API. SSA binds authenticated impressions to secret-shared reports without modifying the underlying MPC aggregation protocol. This illustrates the kind of targeted cryptographic innovation that may strengthen fraud resistance within the emerging standard. The authors recently presented their proposal to the W3C community group connected to Attribution.

Having said that, fraud in advertising ecosystems is a broad and evolving problem that likely extends beyond impression authentication. Future directions may require incorporating carefully designed signals at the accountability layer of Attribution, enabling detection patterns without undermining aggregate privacy guarantees. In particularly sensitive cases, narrowly scoped exceptions – such as special-purpose tracking strictly limited to fraud detection and explicitly prohibited from secondary use – may also be necessary to balance ecosystem integrity with privacy commitments. We urge researchers to consider such broad design spectra.

4.8 Economics and Market Structure

Finally, a crucial research direction concerns Attribution's broader implications for the structure of the advertising ecosystem. Even with technical mitigations aimed at supporting smaller participants, utility is unlikely to degrade

uniformly across actors. This raises concrete questions: How do privacy constraints affect market entry and competition? Does aggregation structurally advantage entities with substantial first-party data? Do cross-querier modes lower or raise barriers to entry? Which API design choices materially influence competitive dynamics? And how do market indicators compare across Attribution-based advertising, tracking-based regimes, and the (likely transitional) period in which both coexist?

A central challenge is how to ground these questions in rigorous analyses and evaluations, rather than keep them at speculative levels. While we are not economics experts, one technical question is how we could measure Attribution's utility relative to tracking-based systems. The comparison is not straightforward. Tracking regimes already operate under incomplete observability due to signal loss, ad blockers, cookie restrictions, and cross-device fragmentation, producing attribution that is often noisy or biased. By contrast, Attribution releases aggregated but structurally precise measurements (barring fraud), with well-defined semantics and controlled noise. It is therefore not self-evident that Attribution is strictly less informative in practice. Careful empirical work is needed to define meaningful utility metrics and to compare signal quality, robustness, and economic outcomes across regimes.

Beyond the empirical question of how Attribution's utility compares to tracking, there is a broader need for rigorous analysis at the intersection of advertising economics, competition policy, and privacy technology. Understanding how privacy-constrained aggregation reshapes incentives, market entry, and competitive dynamics is essential to evaluating Attribution's real-world impact. These studies should assess the limitations of the current design, but also try to identify what architectural or governance changes would be required to support a competitive ecosystem – for example, whether carefully designed observability or transparency signals about API usage could mitigate concentration risks without weakening privacy guarantees.

5 Broader Significance: Toward a Narrow Waist for Privacy-Preserving Aggregation

While this article centers on web advertising measurement, Attribution's successful standardization and deployment could matter well beyond that domain. Aggregation-oriented workloads now dominate many settings – including mobile platforms, smart spaces, and AI systems – yet they are still commonly served through tracking and backend data collection, because platforms lack comparably usable, privacy-preserving alternatives. Attribution offers a rare chance to validate a different design point. But beyond its specific application to web advertising, it raises a broader architectural question that we believe deserves explicit attention from the research community.

5.1 Could an Attribution-Like API Become a Narrow Waist?

The question is this: *could a generalization of Attribution's architecture serve as a common substrate – a narrow waist – for privacy-preserving aggregation across platforms and domains*?

We are not claiming that Attribution itself, as specified for web advertising, is the right design for other settings. The question is whether an *Attribution-like API* – one that preserves the high-level architecture while abstracting away web-specific details – could be broadly applicable. Some architectural commitments of such a generalization could be: on-device IDP privacy budgeting (so the new API can semantically be aligned with the web standard); a simple, (largely) summation-based MPC aggregation layer with DP noise added centrally (so secure aggregation can be done as efficiently

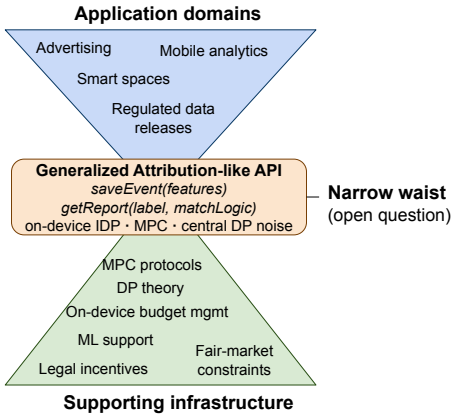

Fig. 3. **Narrow waist effect.**

as possible); and an interface structured around two primitives, `saveEvent(features)`, which registers a local event with associated features on-device, related to some user activity of interest, and `getReport(label, matchLogic)`, which attaches a label to a previously registered event matching some specified logic, and releases an encrypted labeled example for subsequent MPC-based DP aggregation. This architecture appears already general enough to express conversion attribution on the web; the question is whether it can express enough of the aggregation workloads in mobile analytics, smart spaces, regulated data releases, and elsewhere to make a shared implementation worthwhile.

We believe this question is extremely important, and that the research community should engage with it seriously. The reason is the potential for a *narrow waist effect* that might arise if we can find a general API aligned with the cross-company web standard (Fig.3). In network architecture, a narrow waist refers to a thin, stable interface layer that diverse applications above it and diverse infrastructure below it can both depend on, enabling independent innovation on both sides. TCP/IP, standardized by the IETF in the early 1980s, is the canonical example. TCP/IP was not optimal for any particular use case – it was not the fastest, not the most efficient, and not well suited to real-time applications. But its standardization as a common substrate unleashed the Internet: it concentrated infrastructure investment, created a stable target for application developers, and enabled cumulative progress across what had previously been fragmented, incompatible networks.

The analogy is instructive for privacy-preserving aggregation, which today is similarly fragmented. Research and deployment efforts around federated learning, secure aggregation, and differential privacy are scattered across incompatible APIs, threat models, and privacy semantics. This fragmentation diffuses effort: cryptographers optimize MPC protocols for bespoke workloads that cannot be reused; DP theorists develop composition frameworks calibrated to specific deployment assumptions; systems researchers build on-device mechanisms that cannot be shared across platforms. A standardized, Attribution-like aggregation interface – even one that is not optimal for every use case – could change this dynamic by concentrating innovation on both sides of the waist simultaneously (Fig.3). Cryptographers could optimize MPC for a stable, well-defined class of aggregation workloads. DP theorists could focus on composition, bias control, and empirically grounded parameter selection within a shared semantic framework. Systems researchers could build on-device budget management and usability tooling that generalizes across platforms. ML researchers could develop learning algorithms designed for aggregate-only supervision. Legal and economic scholars could evaluate regulatory differentiation grounded in a concrete, widely deployed architecture rather than abstract proposals. And atop this shared infrastructure, applications could flourish: advertising measurement, mobile analytics, physical-space sensing without tracking, privacy-preserving data releases mandated by competition regulators.

Whether an Attribution-like API can genuinely occupy this narrow-waist position remains an open research question. It is possible that the architectural commitments described above are too restrictive for important classes of aggregation workloads, or that different domains require sufficiently different threat models as to preclude meaningful standardization. These are legitimate concerns, however our proposal to the community is to investigate: to characterize which aggregation workloads fit naturally within the Attribution-like architecture, which require extensions, and which are genuinely incompatible, and to design and evaluate those extensions rigorously. Attribution's standardization creates a rare and time-limited window in which this investigation can influence real platform infrastructure. We urge the community to take it.

The remainder of this section describes three settings – mobile operating systems, smart spaces, and first-party data sharing – where investigating this question would be extremely valuable, though we refrain from making claims that the specific Attribution API itself is suitable.

5.2 Mobile Operating Systems: Separating Primary Use from Secondary Aggregation

Mobile operating systems mediate some of the most sensitive and frequently accessed user signals, including location, motion sensors, microphones, and app interaction events. These signals are essential for *primary functionality* – navigation, ride sharing, media playback – which often requires short-lived, user-visible access to raw data. At the same time, the *same signals* increasingly power *secondary uses*, which are often aggregation-oriented: analytics, recommendation, advertising optimization, and model training.

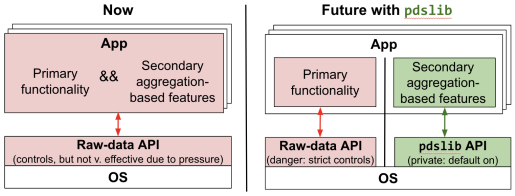

Fig. 4. **Separating primary functionality from secondary aggregation via OS-managed aggregation.**

Today, these two uses are typically coalesced within applications and exposed through the same raw-data APIs and permission mechanisms (Fig.4, left side). As a result, applications that legitimately need short-lived access for primary functionality often request (and sometimes obtain) broad or persistent permissions (e.g., background location) to support secondary aggregation. Tutorials explicitly advising developers how to elicit such permissions exist [32], and prior work shows that pervasive tracking, opaque data flows, and widespread leakage arise via background components and third-party SDKs [22, 28, 31, 34, 41]. From the platform's perspective, this creates a persistent tension: permission prompts and controls attempt to constrain behavior that the API itself structurally encourages.

Attribution suggests a different design point. Imagine it were to be generalized into an *OS-level privacy-preserving aggregation primitive* that worked well for mobile use cases. Then, by exposing aggregation as a first-order primitive, *enabled by default* with strong OS-enforced privacy parameters, the mobile OS can route secondary aggregation workloads through this interface while reserving raw-data access for primary functionality only (Fig.4, right side). For example, an app that today combines sporadic foreground location access (e.g., navigation) with background, location-intensive aggregation (e.g., learning which locations drive engagement) could relinquish persistent background permissions and rely instead on OS-managed aggregation. Developers gain a cleaner, more reliable mechanism, while users avoid pervasive collection by myriad apps.

Critically, the efforts on Attribution surface what such a primitive must handle to be viable at scale: on-device privacy accounting, contention among many competing applications, resilience to depletion and abuse, and observability without raw-data access. These challenges do not disappear at the OS layer; they become more acute. Attribution can provide a blueprint for how those might be addressed – and potentially a springboard for their adoption, since W3C standards often impact mobile OS processes even though technically they only govern web interactions.

5.3 Smart Spaces: From Web Privacy to the Physical World

A similar aggregation-versus-tracking tension is emerging in smart spaces, where sensing technologies capture movement, presence, and interaction in public environments. Large-scale deployments have sometimes faced public opposition – as in the Toronto Sidewalk Labs project [36] – yet smaller roll-outs continue to appear with varying degrees of transparency and data-management rigor. In New York City, for example, CitiBike publishes "anonymized" ride datasets with precise timestamps and coordinates, enabling trajectory reconstruction despite the absence of explicit identifiers

(as our own research has shown); public camera feeds are widely accessible; LinkNYC kiosks have reportedly collected device identifiers and are equipped with dormant cameras that raise concerns [25]; and Wegmans has disclosed (as required by law) the use of facial recognition in stores without detailing its data collection and retention practices [27].

We do not suggest that technical standards alone can resolve broader political questions surrounding monitoring of public space. However, if urban sensing expands – even incrementally – the absence of clear data-access and management standards risks repeating the web's trajectory. Cross-site tracking was allowed to proliferate for decades before browsers introduced structured, privacy-preserving alternatives such as Attribution. As a result, privacy-preserving aggregation must now compete with an entrenched tracking ecosystem, complicating its adoption. Smart spaces, by contrast, remain comparatively early in their deployment. If society permits certain sensing uses – for safety, infrastructure planning, or accessibility – then defining structured access models *before* backend data collection and cross-context tracking become the default may be critical.

What might such models look like? The web provides a useful reference framework, with Attribution offering a concrete architectural component. Although the web governs virtual activity and smart spaces govern physical movement, their data-access patterns are closely analogous. In both domains, we can distinguish at least three levels of access, each with distinct privacy implications.

First, *ephemeral, in-context use*. On the web, this corresponds to first-party, in-session access for immediate functionality. In smart spaces, the analog is on-scene sensor use that supports real-time tasks – such as detecting pedestrians for traffic control – without retaining or redistributing raw trajectories. Public norms often tolerate this level of access when it does not persist.

Second, *bounded, longitudinal aggregation*. On the web, this resembles first-party collection within a single site over time. In urban contexts, it corresponds to aggregation within a defined locality – for example, counting pedestrians at an intersection across weeks to guide planning. Even here, persistent recording can raise concerns, motivating aggregation and differential privacy as safeguards.

Third, *cross-context aggregation*. On the web, this is the most contested form of tracking: linking activity across sites via third-party cookies or fingerprinting. Attribution seeks to replace that model with a structured, budgeted aggregation interface enforced by the platform. In smart spaces, the analog is cross-locality tracking of individuals as they move across neighborhoods – for example, reconstructing trajectories across multiple sensing points. Such measurements are most closely associated with surveillance and public backlash. A generalized form of Attribution could instead enable cross-locality *measurement* without enabling cross-locality *tracking*.

The architectural lesson is that not all sensing requires centralized raw-data retention; many aggregate use cases may not require it at all if the interface is structured appropriately. Attribution demonstrates how cross-context measurement can be mediated through platform-managed aggregation under enforceable privacy semantics. While smart-space deployments raise broader societal questions – and physical-world sensing carries higher stakes because it cannot be avoided by simply leaving a device behind – the web experience suggests concrete design principles: separate ephemeral and localized use from longitudinal analytics, and replace cross-context tracking with structured, privacy-bounded interfaces.

We are currently exploring a privacy architecture for smart cities rooted in these principles, to test whether many valuable use cases can in fact be supported without centralized raw-data retention.

5.4 Standardized, Privacy-Preserving First-Party Data Sharing

Beyond cross-site advertising measurement, the generalized API described in §5.1 points toward a broader architectural problem: how platforms could support the sharing of first-party data under enforceable privacy guarantees. Today, first-party data is routinely shared through backend exchanges, including commercial data sales and partnerships that are often opaque to users. In addition, regulatory mandates may require first-party data releases, as in the recent requirement that Google Search provide user-derived datasets – such as search-click data – to promote competition [11]. In the absence of standardized technical mechanisms for privacy-preserving release at this scale, such transfers rely on bespoke solutions that must confront difficult privacy-utility tradeoffs, compounded by the large-scale nature of these releases.

This setting naturally raises the question of whether Attribution, or an adapted form of it, could support structured sharing of first-party data with external parties – at least in cases where aggregate statistics suffice. Sharing first-party data with other entities functionally resembles cross-site tracking, except that it occurs via backend data flows rather than through the browser. An Attribution-like approach would shift this model: external recipients could receive encrypted reports and compute aggregate measurements under browser-enforced privacy budgets, rather than obtaining raw logs. We do not claim that such an approach is technically feasible or legally appropriate. However, the broader research question is interesting: can backend first-party data exchanges be replaced, in some useful settings, with browser-mediated, privacy-bounded aggregation interfaces that enforce uniform privacy semantics?

In this sense, Attribution's significance extends beyond advertising. If developed to maturity, it could provide a shared architectural foundation for exposing sensitive, population-level data under enforceable privacy semantics – reducing fragmentation, discouraging ad hoc raw-data exchange, and avoiding the repeated reinvention of privacy mechanisms across domains.

6 Summary and Call for Action

If successful, Attribution will be the first instance of a *platform-managed, privacy-preserving aggregation API* jointly adopted across multiple companies and deployed on billions of devices. It will demonstrate that aggregation – rather than raw data access – can be elevated to a first-class platform primitive. Its architectural principles offer a blueprint for other domains: a structured aggregation interface; on-device privacy enforcement and budgeting; global privacy semantics enforced at the platform layer; and a clear separation between functionality that can be supported privately and functionality that cannot.

We are therefore at a consequential moment. Attribution is under standardization, and key aspects of its design, parameters, and infrastructure remain unsettled. Decisions made now will determine whether privacy-preserving aggregation becomes a durable platform foundation – or a narrow abstraction that coexists with tracking without displacing it.

The gaps are real. Achieving high utility under strong privacy, supporting actors of different scales, enabling accountability without raw-data visibility, and scaling secure aggregation economically are unsolved technical challenges. These are not incremental refinements; they are foundational systems and theory problems.

We urge the research community to treat Attribution not as an endpoint, but as a blueprint: to stress-test it, extend it, generalize it, and refine its components while preserving its core commitment to rigorous, enforceable privacy. Done well, this effort can shape not only the future of web advertising, but a broader class of privacy-preserving data-access APIs across platforms.

This is a rare opportunity to influence privacy infrastructure before it hardens. We invite the community to engage with focus and determination.

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